



National Forensics Sciences University, Goa Campus

Mid- semester Examination

Programme – M.Sc. Digital Forensics and Information Security		Sem – II														
Date – 19/03/2026																
Subject Name: Artificial Intelligence		Subject Code - CTMSDFIS SII P4														
Time- 1.5 Hours		Max.Marks - 50														
Instructions - 1) Answer all questions. 2) Assume suitable data.																
Q.1	Answer any four	20 marks														
	a. Define Mean, Median, and Mode. Explain a scenario where the Median would be a more reliable measure of central tendency than the Mean.	5 marks														
	b. Compare and contrast Conditional Statements (if-else) and Loops (for/while) in Python. Provide a brief code snippet demonstrating how to use a loop to iterate through a List.	5 marks														
	c. Explain the primary differences between a Python List and a NumPy Array. In what scenarios would you prefer one over the other for ML tasks?	5 marks														
	d. Differentiate between Supervised and Unsupervised Learning. Provide two practical application examples for each.	5 marks														
	e. Describe the processes of Feature Selection and Feature Normalization. Why are these steps critical before feeding data into a Machine Learning model?	5 marks														
Q.2	Attempt all	15 marks														
	a. Explain the difference between Correlation and Regression. Does a high correlation between two variables imply that one causes the other? Justify your answer.	5 marks														
	b. Explain the concepts of Training, Testing, and Validation sets. Why is it considered bad practice to evaluate a model's final performance using the Training set?	5 marks														
	c. Describe Underfitting and Overfitting using the concepts of Bias and Variance. How can Hyperparameters be used to mitigate overfitting?	5 marks														
Q. 3	Attempt a and b	15 marks														
Q.3 a	Attempt any one															
Q.3 a	I. Develop a Linear Regression Model using the below data to predict Total Salary. Find the Slope and Intercept of the Equation. <table><tr><th>Projects Completed (x)</th><th>Total Salary in Thousands (y)</th></tr><tr><td>2</td><td>60</td></tr><tr><td>4</td><td>70</td></tr><tr><td>6</td><td>80</td></tr><tr><td>8</td><td>90</td></tr><tr><td>10</td><td>100</td></tr><tr><td>12</td><td>110</td></tr></table>	Projects Completed (x)	Total Salary in Thousands (y)	2	60	4	70	6	80	8	90	10	100	12	110	8 marks $m = 1.22$ $b = 76.46$ $y = 127.7$
Projects Completed (x)	Total Salary in Thousands (y)															
2	60															
4	70															
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12	110															

	OR	
	II. Critically analyze the relationship between abstraction and generalization in Machine Learning. Explain both these concepts with respect to Machine Learning.	8 marks
Q.3 b	Attempt any one	7 marks
Q.3 b	I. A high-security airport terminal implemented a new AI-driven baggage scanner to detect prohibited items. Out of 1,000 suitcases screened, 80 bags actually contained prohibited items, while 920 bags were perfectly clear. The scanner successfully flagged 72 of the prohibited bags, but it also incorrectly triggered an alarm for 46 clear bags, causing them to be pulled aside for manual inspection. Construct the confusion matrix, calculate the Recall, and explain what the False Negative (FN) value represents in the context of airport safety	7 marks FN=8 Recall = 0.99, 99.1%
	OR	
	II. Explain the conceptual framework of Reinforcement Learning (RL). Using a suitable block diagram.	7 marks

*** All the best***